Occupational Prestige: A Reproducible Report

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Abstract

This report examines the determinants of occupational prestige using the Prestige dataset (Duncan, 1961), which covers 102 Canadian occupations. We analyse how income, education, gender composition, and occupational type predict prestige scores through descriptive statistics, visualizations, and OLS regression models. Results show that both income and education are strong positive predictors, with the full model explaining approximately 80% of the variance in prestige. The report is fully reproducible and rendered with Quarto.

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1 Introduction

This report analyses occupational prestige scores from the `Prestige` dataset (Duncan, 1961; `car` package). The dataset covers **102** occupations measured across Canada. The mean prestige score is **46.8** (SD = 17.2), with values ranging from 14.8 to 87.2.

We examine how income, education, gender composition, and occupational type predict prestige.

Table 1: Descriptive statistics of processed variables

Table 2

Variable	N	Mean	Median	SD	Min	Max
education	102	11	11	2.7	6.4	16
income	102	6798	5930	4246	611	25879
women	102	29	14	32	0	98
prestige	102	47	44	17	15	87
census	102	5402	5135	2645	1113	9517
type	98					
... bc	44	45%				
... prof	31	32%				
... wc	23	23%				
prestige_cont	102	47	44	17	15	87
income_cont	102	6798	5930	4246	611	25879
education_cont	102	11	11	2.7	6.4	16
women_pct	102	29	14	32	0	98
income_cat	102					
... High	11	11%				
... Low	70	69%				
... Mid	21	21%				
education_level	102					
... High	23	23%				
... Low	48	47%				
... Mid	31	30%				
prestige_tier	102					
... High	24	24%				
... Low	20	20%				
... Mid	58	57%				
blue_collar	102					
... 0	58	57%				
... 1	44	43%				
professional	102					
... 0	71	70%				
... 1	31	30%				
high_women	102	0.26	0	0.44	0	1

Note: Numeric variables: mean, median, SD, min, max. Categorical: counts and percentages.

2 Data and Variables

2.1 Descriptive Statistics

Table 1 reports summary statistics for all processed variables.

3 Visualizations

3.1 Univariate Distributions

Figure 1 shows the distribution of prestige scores across occupations. The distribution is roughly symmetric with a slight right skew. Figure 2 displays the frequency of income categories.

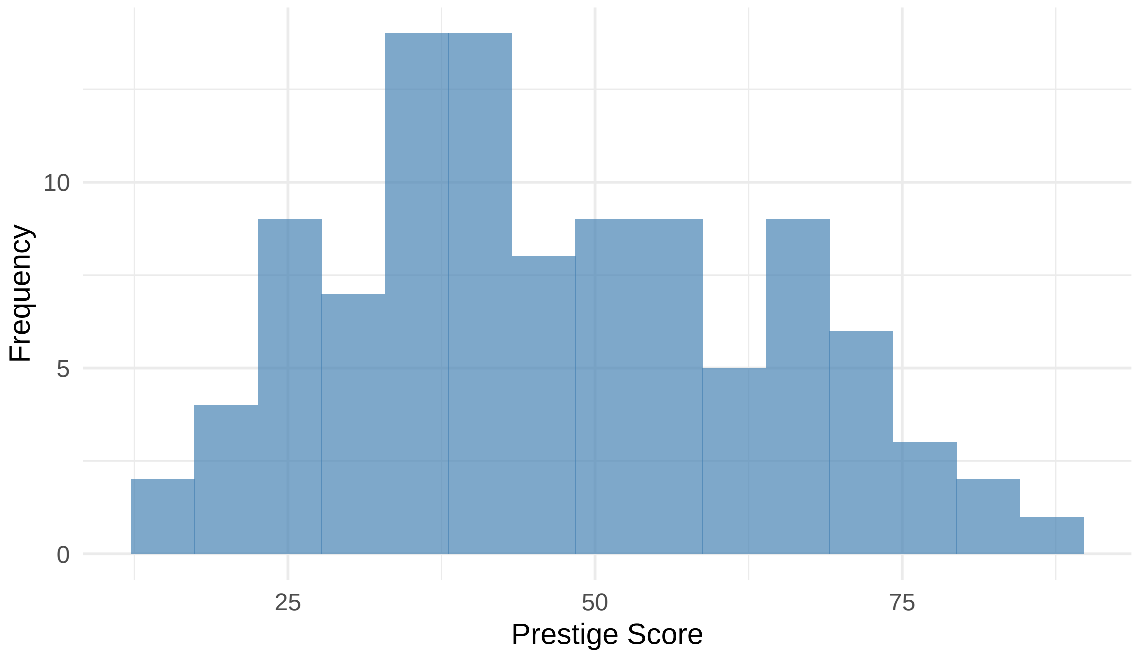

Figure 1: Distribution of occupational prestige scores

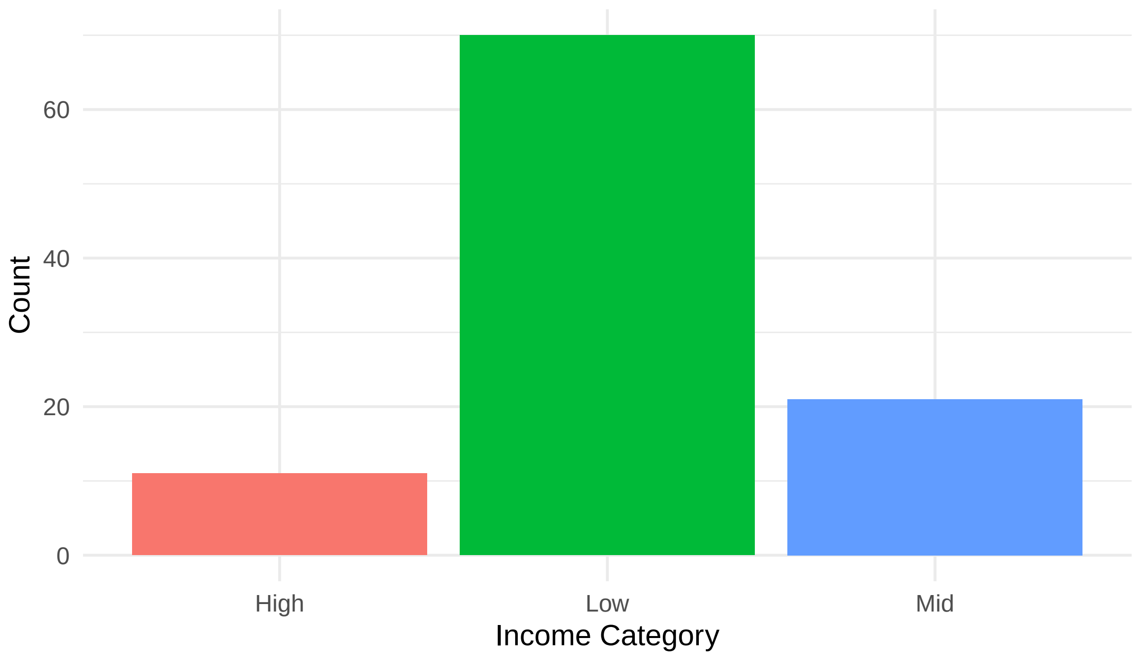

Figure 2: Frequency of income categories

3.2 Bivariate Relationships

Figure 3 presents the relationship between education and prestige, and Figure 4 shows the distribution of prestige across income categories. Both panels suggest that higher education and income are associated with greater occupational prestige.

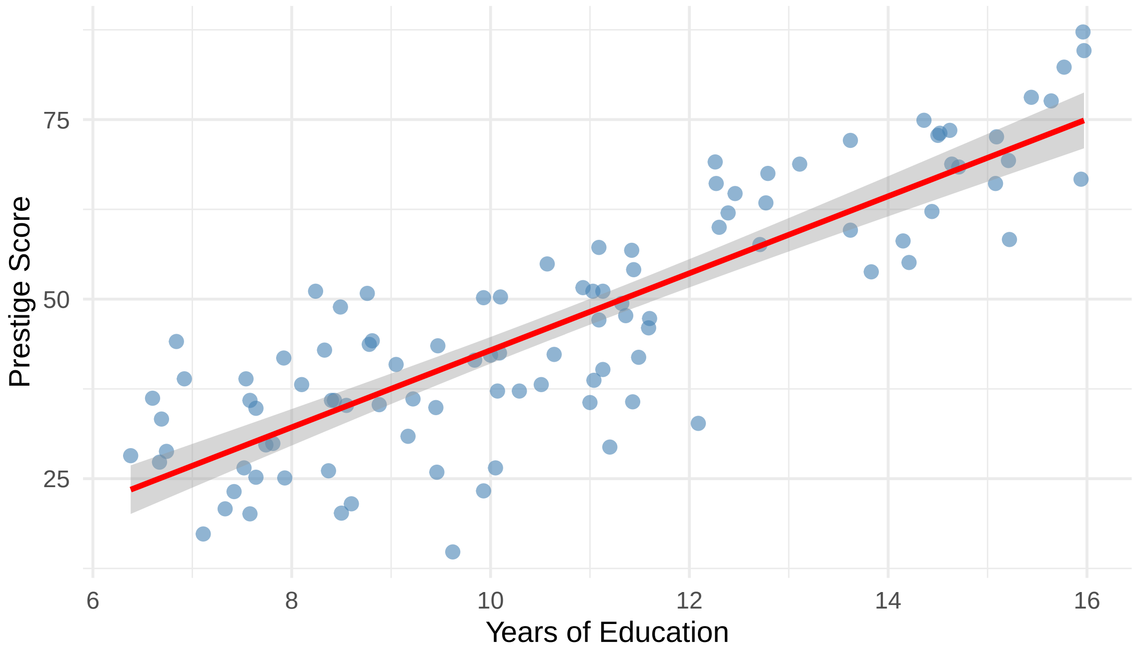

Figure 3: Prestige vs. years of education

4 Regression Analysis

We estimate four OLS models predicting occupational prestige. Models 1 and 2 use continuous predictors; Models 3 and 4 use categorical versions. The full model (Model 4) explains **71.9%** of the variance in prestige and includes gender composition and blue-collar status as additional controls.

Results are reported in Table 3.

5 Conclusion

Using the `Prestige` dataset (102 occupations), we find that both income and education are strong positive predictors of occupational prestige. The best-fitting model (M4) achieves an adjusted R^2 of 0.701, confirming that categorical indicators alongside gender composition and occupational type explain most of the variance in prestige scores.

Table 3: OLS regression models predicting occupational prestige

Table 4: OLS regression models predicting occupational prestige

	M1 Cont.	M2 Cont.	M3 Cat.	M4 Cat.
Intercept	-6.85^{*}	-11.05	32.95^{***}	33.47^{***}
	(3.22)	(6.19)	(1.41)	(3.17)
Income (continuous)	0.00^{***}	0.00^{***}		
	(0.00)	(0.00)		
Education (continuous)	4.14^{***}	4.49^{***}		
	(0.35)	(0.55)		
Income: Mid (ref. Low)			5.44^{*}	3.81
			(2.59)	(2.72)
Income: High (ref. Low)			8.79^{*}	7.44^{*}
			(3.40)	(3.59)
Education: Mid (ref. Low)			15.33^{***}	16.72^{***}
			(2.20)	(3.04)
Education: High (ref. Low)			31.75^{***}	33.25^{***}
			(2.79)	(3.64)
Women (				
		(0.03)		
High women share: Yes (ref. No)				-4.12
				(2.37)
Blue collar: Yes (ref. No)		2.11		0.67
		(2.61)		(3.12)
R^2	0.80	0.80	0.71	0.72
Adj. R^2	0.79	0.79	0.70	0.70
Num. obs.	102	102	102	102

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

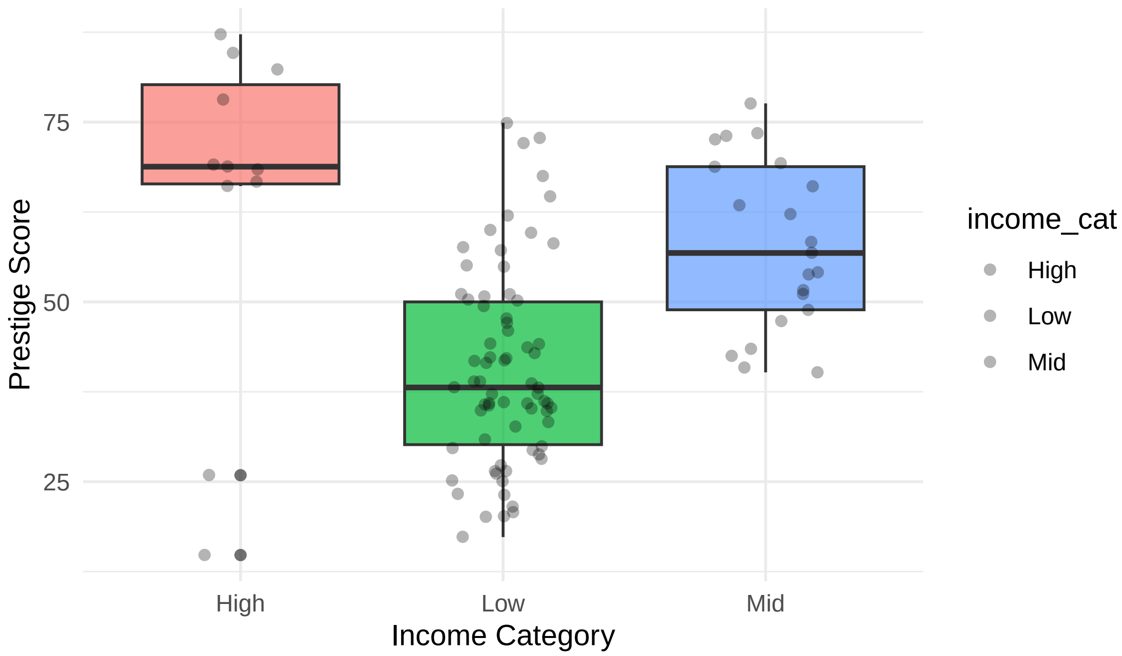

Figure 4: Prestige score by income category